S1 — Foundations of Chemistry

1.1 Atomic Structure

The Atom

Atom: The smallest particle of an element that can take part in a chemical reaction. Atoms consist of a small, dense, positively charged nucleus containing protons and neutrons, surrounded by electrons in energy levels (shells).

Particle	Relative Mass	Relative Charge	Location
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	1/1836 ≈ 0	−1	Shells / orbitals

The **atomic number (Z)** is the number of protons in the nucleus and defines the element. The **mass number (A)** is the total number of protons and neutrons. For a neutral atom the number of electrons equals the number of protons.

Isotopes

Isotopes: Atoms of the same element (same number of protons) with different numbers of neutrons and therefore different mass numbers. Isotopes have identical chemical properties because they have the same electron configuration, but different physical properties (e.g. density, rate of diffusion).

Examples: carbon-12 (^{12}C, 6p + 6n) and carbon-13 (^{13}C, 6p + 7n). The existence of isotopes means most elements do not have whole-number relative atomic masses.

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Mass Spectrometry

A mass spectrometer measures the relative masses and abundances of isotopes or molecules. The basic steps are:

1. **Ionisation:** The sample is vaporised and bombarded with high-energy electrons to form positive ions (electron impact ionisation) — an electron is knocked off each atom/molecule.
2. **Acceleration:** Ions are accelerated by an electric field so they all have the same kinetic energy.
3. **Deflection:** Ions pass through a magnetic field. Lighter ions and ions with higher charge are deflected more.
4. **Detection:** Ions reach the detector, producing a current proportional to their abundance.

The mass spectrum shows peaks at each m/z value. The relative atomic mass (A_r) of an element is calculated from the mass spectrum using:

$$A_r = \Sigma(\text{isotope mass} \times \%\ \text{abundance})\ /\ 100$$

Exam Tip: In mass spectra of molecules, the peak with the highest m/z value (ignoring the M+1 peak) is the molecular ion peak (M^+), giving the relative molecular mass. Fragmentation patterns help identify functional groups and structural features.

Worked Example — Calculating A_r:

Chlorine has two isotopes: ^{35}Cl (abundance 75.8%) and ^{37}Cl (abundance 24.2%).

A_r = (35 × 75.8 + 37 × 24.2) / 100 = (2653.0 + 895.4) / 100 = **35.5**

Electron Configuration

Electrons occupy atomic orbitals. An orbital is a region of space around the nucleus where there is a high probability of finding an electron. Each orbital holds a maximum of two electrons with opposite spins (Pauli exclusion principle).

Sub-shell	Number of orbitals	Max electrons	Shape

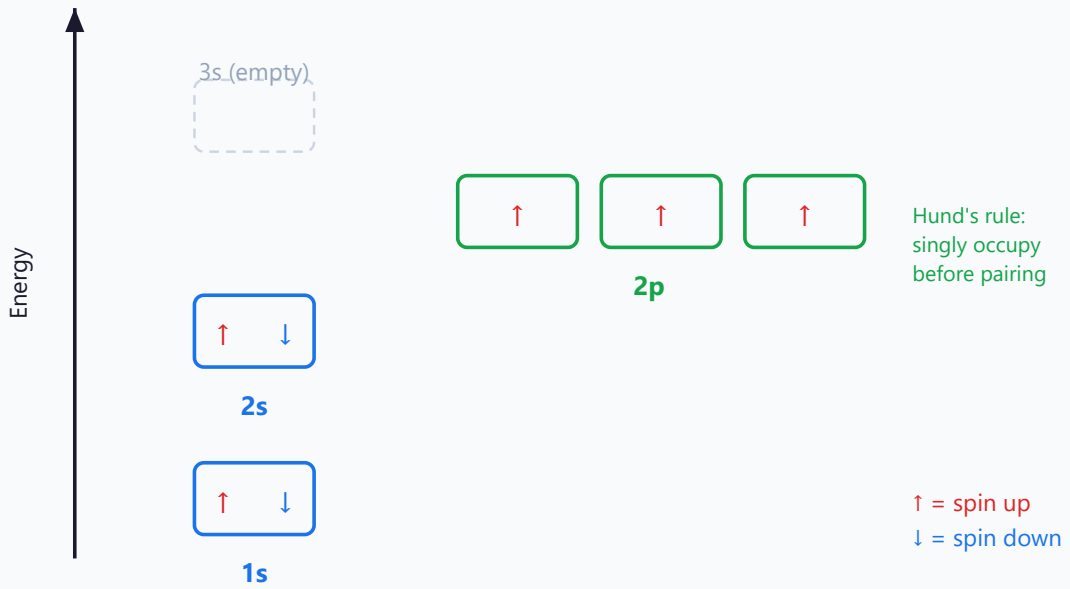
s	1	2	Spherical
p	3	6	Dumbbell
d	5	10	Complex / cloverleaf
f	7	14	Complex

The order of filling follows the Aufbau principle: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p ... Note that 4s fills before 3d. When writing electron configurations for ions, remove electrons from the highest principal energy level first (e.g. Fe: [Ar] $3d^6\ 4s^2$, but Fe^{2+}: [Ar] $3d^6$).

> **Hund's rule:** Electrons occupy orbitals of the same sub-shell singly before pairing up, and with parallel spins. This minimises electron-electron repulsion.

Chromium and copper are exceptions: Cr is [Ar] $3d^5\ 4s^1$ (not $3d^4\ 4s^2$) and Cu is [Ar] $3d^{10}\ 4s^1$ (not $3d^9\ 4s^2$) because half-filled and fully-filled d sub-shells have extra stability due to symmetrical electron distribution and favourable exchange energy.

Electron Configuration — Orbital Box Diagram for Nitrogen ($1s^2\ 2s^2\ 2p^3$)



Ionisation Energies

First ionisation energy: The energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous 1+ ions.
$X(g) \rightarrow X^{+}(g) + e^{-}$

Ionisation energy depends on: nuclear charge, atomic radius, and shielding (screening) by inner electrons.

Trends:

- **Across a period:** Generally increases because nuclear charge increases while shielding remains roughly constant, so the outer electron is held more tightly.
- **Down a group:** Decreases because the atomic radius increases and shielding increases, outweighing the increase in nuclear charge.

There are two anomalies across Period 3: a dip from Mg to Al (the 3p electron in Al is easier to remove than a 3s electron in Mg because the 3p sub-shell is higher in energy and slightly shielded by 3s) and a dip from P to S (the paired 3p electron in S experiences extra repulsion, making it easier to remove).

Exam Tip: Successive ionisation energies can reveal the number of electrons in each shell. A large jump between successive values indicates that the next electron is being removed from a shell closer to the nucleus (less shielded, more tightly held).

1.2 Amount of Substance

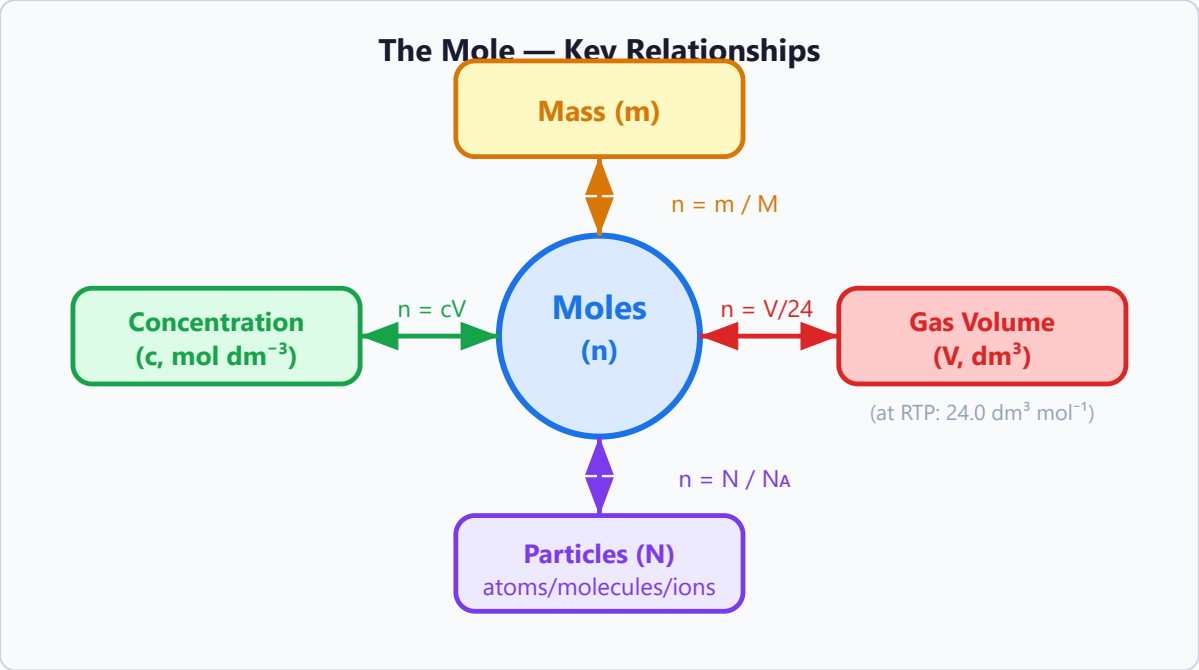
The Mole and Avogadro's Constant

Mole: The amount of substance that contains 6.022×10^{23} particles (atoms, molecules, ions, etc.). This number is Avogadro's constant (N_A).

Molar mass (M): The mass of one mole of a substance, in g mol^{-1}. Numerically equal to the relative formula mass.

$$n = m / M$$

where n is the number of moles, m is the mass in grams, and M is the molar mass in g mol^{-1}.



Empirical and Molecular Formulae

Empirical formula: The simplest whole-number ratio of atoms of each element in a compound.

Molecular formula: The actual number of atoms of each element in one molecule of a compound.

To find the empirical formula from percentage composition: divide each percentage by the element's A_r, then divide all results by the smallest value to get the simplest whole-number ratio. The molecular formula is a whole-number multiple of the empirical formula: molecular formula = (empirical formula) × n, where n = M_r / empirical formula mass.

Finding Empirical & Molecular Formulae

Step 1 Mass or % of each element → **Step 2** Divide each by its A_r → moles → **Step 3** Divide all by the smallest value → **Empirical Formula** (simplest ratio)

Worked Example: Compound with 40.0% C, 6.7% H, 53.3% O

Element	% / A_r	Moles	Ratio
C	40.0 / 12.0	3.33	**1**
H	6.7 / 1.0	6.70	**2**
O	53.3 / 16.0	3.33	**1**

Empirical formula = CH_2O

If M_r = 60: n = 60/30 = 2, so molecular formula = $C_2H_4O_2$ (ethanoic acid)

Molar Gas Volume

At room temperature and pressure (RTP, approximately 25 °C and 1 atm), one mole of any gas occupies approximately 24.0 dm^3 (24,000 cm^3).

$$n = V / V_m$$

where V is the volume in dm^3 and V_m = 24.0 dm^3 mol^{-1} at RTP.

The ideal gas equation provides a more general relationship:

$$pV = nRT$$

where p is pressure in Pa, V is volume in m^3, n is moles, R = 8.314 J K^{-1} mol^{-1}, and T is temperature in K.

Concentrations of Solutions

$$c = n / V$$

where c is concentration in mol dm^{-3}, n is moles of solute, and V is volume of solution in dm^3. To convert cm^3 to dm^3, divide by 1000.

Exam Tip: Always check your units. A very common error is using volume in cm^3 instead of dm^3 in the concentration formula. Remember: 1 dm^3 = 1000 cm^3. For pV = nRT, pressure must be in Pa, volume in m^3, and temperature in K.

Worked Example — Titration Calculation:

25.0 cm^3 of Na_2CO_3 solution of unknown concentration was titrated with 0.100 mol dm^{-3} HCl. The mean titre was 27.60 cm^3.

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$

n(HCl) = 0.100 × 27.60/1000 = 2.76×10^{-3} mol

n(Na_2CO_3) = 2.76×10^{-3} / 2 = 1.38×10^{-3} mol (1:2 ratio)

[Na_2CO_3] = 1.38×10^{-3} / 0.0250 = **0.0552 mol dm^{-3}**

Balanced Equations and Reacting Masses

A balanced chemical equation shows the molar ratios of reactants and products. Use these ratios to calculate reacting masses, volumes of gases, and volumes of solutions. The general approach:

1. Write the balanced equation.
2. Calculate moles of the known substance.
3. Use the molar ratio to find moles of the required substance.
4. Convert moles to the desired quantity (mass, volume, concentration).

Percentage Yield and Atom Economy

$$\text{Percentage yield} = (\text{actual yield} / \text{theoretical yield}) \times 100\%$$

$$\text{Atom economy} = (M_r \text{ of desired product} / \Sigma\, M_r \text{ of all products}) \times 100\%$$

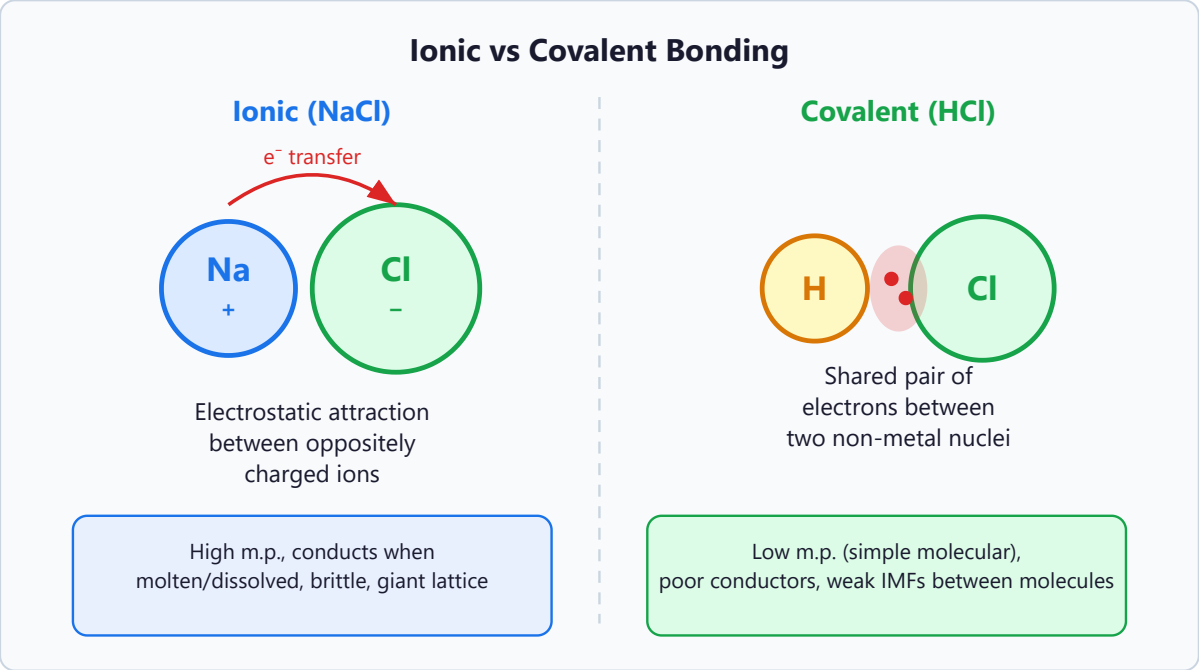
A high atom economy means less waste. Addition reactions have 100% atom economy. Substitution and elimination reactions have lower atom economies because by-products are formed.

1.3 Bonding and Structure

Ionic Bonding

Ionic bonding: The electrostatic attraction between oppositely charged ions, formed by the transfer of electrons from a metal to a non-metal. Metals lose electrons to form cations; non-metals gain electrons to form anions.

Ionic compounds form giant ionic lattices (e.g. NaCl). Properties: high melting points (strong electrostatic forces throughout the lattice), conduct electricity when molten or dissolved (ions are free to move), brittle (displacement of layers causes like charges to align and repel), and dissolve in polar solvents such as water.



Covalent Bonding

Covalent bond: A shared pair of electrons between two atoms, where both nuclei are attracted to the shared electron density. This occurs between non-metal atoms.

Dative (coordinate) bond: A covalent bond in which both electrons in the shared pair come from the same atom (the donor). Represented by an arrow from donor to acceptor. Example: NH_4^+ (the lone pair on nitrogen is donated to H^+).

A **single bond** consists of one shared pair (σ bond). A **double bond** consists of two shared pairs (one σ + one π bond). A **triple bond** consists of three shared pairs (one σ + two π bonds).

Shapes of Molecules — VSEPR Theory

VSEPR (Valence Shell Electron Pair Repulsion) theory: Electron pairs around a central atom arrange themselves to be as far apart as possible to minimise repulsion. Lone pairs repel more than bonding pairs: lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair.

Electron pairs	Bonding pairs	Lone pairs	Shape	Bond angle	Example
2	2	0	Linear	180°	CO_2, $BeCl_2$
3	3	0	Trigonal planar	120°	BF_3, $AlCl_3$
3	2	1	Bent / V-shape	~117°	SO_2
4	4	0	Tetrahedral	109.5°	CH_4, NH_4^+
4	3	1	Trigonal pyramidal	107°	NH_3
4	2	2	Bent / V-shape	104.5°	H_2O
5	5	0	Trigonal bipyramidal	90° & 120°	PCl_5
6	6	0	Octahedral	90°	SF_6

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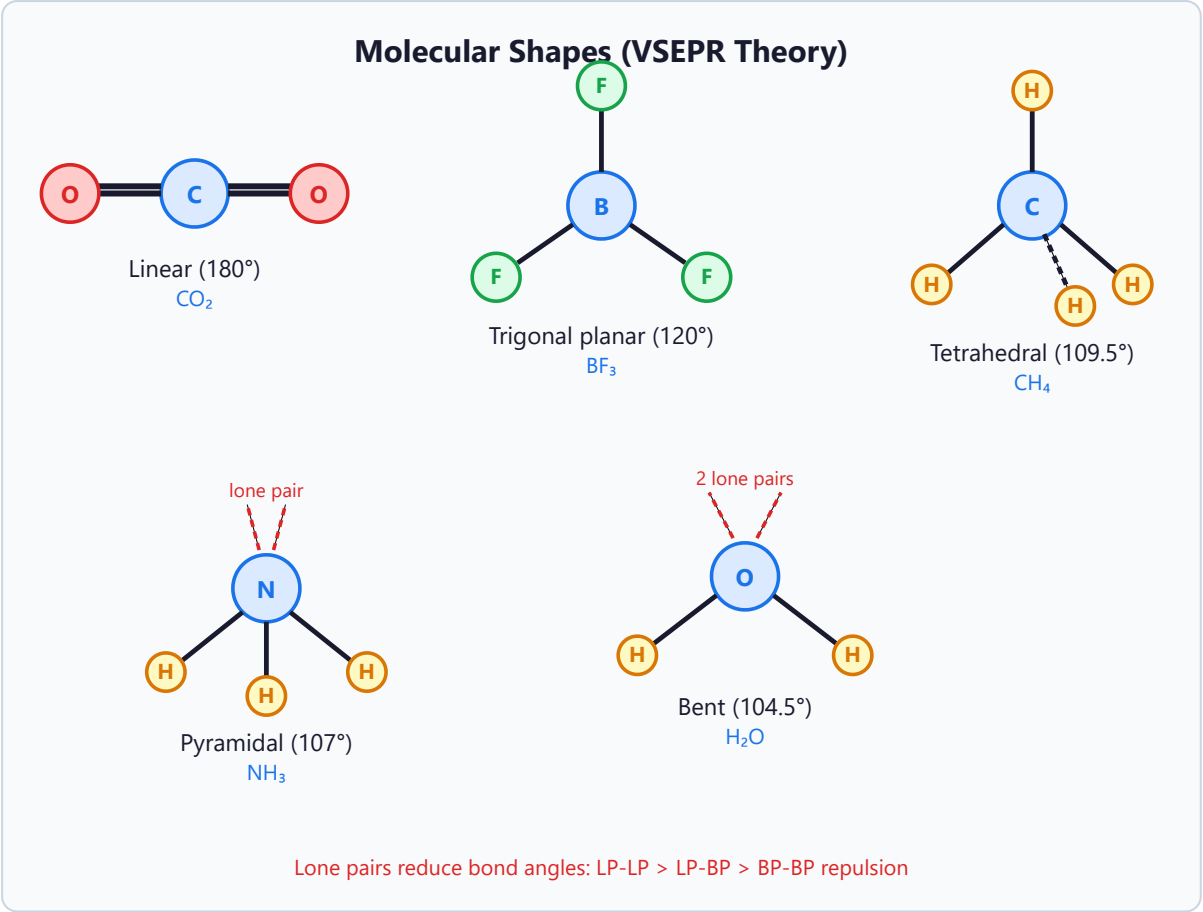

Electronegativity and Polarity

Electronegativity: The ability of an atom to attract the bonding pair of electrons in a covalent bond towards itself. Measured on the Pauling scale. Fluorine is the most electronegative element (4.0).

Trends: Electronegativity increases across a period (greater nuclear charge, smaller atomic radius) and decreases down a group (greater atomic radius, more shielding).

A **polar bond** exists when there is a difference in electronegativity between two bonded atoms. The more electronegative atom carries a partial negative charge (δ−) and the less electronegative atom carries a partial positive charge (δ+). A molecule is polar overall only if the individual bond dipoles do not cancel (i.e. the molecule is not symmetrical). For example, CO_2 has polar bonds but is a non-polar molecule (linear, dipoles cancel), whereas H_2O is a polar molecule (bent shape, dipoles do not cancel).

Metallic Bonding

Metallic bonding: The electrostatic attraction between a lattice of positive metal ions and a sea of delocalised electrons. The outer-shell electrons are free to move throughout the structure.

Properties of metals: high melting points (strong metallic bonds), good electrical conductors (delocalised electrons carry charge), good thermal conductors (delocalised electrons transfer kinetic energy), malleable and ductile (layers of ions can slide over each other without breaking the bonding).

Intermolecular Forces

Intermolecular forces are weak forces between molecules. They determine physical properties such as boiling point and solubility.

Type	Strength	Origin	Present in
London (dispersion) forces	Weakest	Temporary instantaneous dipoles caused by random electron movement induce dipoles in neighbouring molecules	All molecules
Permanent dipole–dipole	Moderate	Attraction between the δ+ end of one polar molecule and the δ− end of another	Polar molecules
Hydrogen bonding	Strongest intermolecular force	A special, strong dipole–dipole interaction between a lone pair on N, O, or F and a hydrogen atom bonded to N, O, or F	NH_3, H_2O, HF, alcohols, carboxylic acids

London forces increase with the number of electrons (and therefore molecular size) because larger electron clouds are more easily polarised, creating stronger temporary dipoles.

Exam Tip: When comparing boiling points, always identify the strongest type of intermolecular force present. Hydrogen bonding explains why water (M_r = 18) has a much higher boiling point than methane (M_r = 16) despite similar molecular masses.

Giant Structures

Giant ionic lattices (e.g. NaCl): Strong electrostatic forces between ions in all directions → high melting points, conduct when molten/dissolved.

Giant covalent (macromolecular) structures (e.g. diamond, silicon dioxide, graphite): Many strong covalent bonds throughout the structure → very high melting points, generally insoluble, generally do not conduct electricity. Exception: graphite has delocalised electrons between layers, so it conducts electricity; the weak London forces between layers make it soft and slippery.

Simple molecular structures (e.g. I_2, H_2O, CO_2): Strong covalent bonds within molecules but weak intermolecular forces between molecules → low melting/boiling points, do not conduct electricity (no charged particles free to move).

Exam Tip: When explaining high melting points, always state the type of bonding/force that must be overcome and describe it as strong. For simple molecular substances, make clear it is the intermolecular forces (not the covalent bonds) that break during melting and boiling. Never say "bonds are broken" when discussing boiling — use "intermolecular forces are overcome."

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